

# Convergence of iterated SOCP refinement for CAZAC feasibility

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## Abstract

Constant-amplitude zero-autocorrelation (CAZAC) sequences satisfy a constant-modulus time condition and a flat-spectrum frequency condition simultaneously — a nonconvex intersection of two tori. Relaxing the equality constraints to convex inequalities gives a compact convex body  $C_\tau$  whose maximal-norm points, at  $\tau = 1$ , are exactly the CAZAC systems, and the natural algorithm iterates a second-order cone program (SOCP) whose objective direction resets to the previous optimizer. The central technical content is Theorem D: a regularity hypothesis (R) needed for a sharp local convergence rate is pinned down exactly at every Zadoff–Chu tuple and length  $n$  (both parities; even  $n$  was previously open), via a mechanism new to this problem — the DFT of a Zadoff–Chu chirp is again a chirp, collapsing a rank computation to counting solutions of  $t^2 \equiv 0 \pmod n$  — so (R) holds iff  $n$  is squarefree. The mechanism is a reusable schema: a different closure fact (self-duality of the Legendre symbol) extends it to a partial regularity result at Björck’s quadratic-phase construction, reduced to one unproved, numerically-confirmed (44 primes) determinant lemma; the same specific mechanism, by contrast, provably cannot reach cubic or higher-degree phase (this does not rule out some other route). This underwrites Theorem C (conditional on (R); unconditional when Theorem A’s limit lies on a Zadoff–Chu tuple’s symmetry orbit and  $n$  is squarefree): near a regular CAZAC point the constraint slack grows *linearly*, not quadratically (KL exponent exactly 0), and the iteration reaches the point exactly after finitely many steps. The theory rests on Theorem A (unconditional): every measurable selection converges to a single fixed point, via an Attouch–Bolte–Svaiter Kurdyka–Łojasiewicz descent argument applied to a set-valued argmax update — a known template, of which Akgas and Kroer [4] (2025, prior art) give a general instance for arbitrary strongly convex objectives; our contribution is the reduction of CAZAC feasibility to  $\max_{C_\tau} \|x\|^2$  (Prop. 2.2) that makes the template apply, not the descent argument itself. This reduction is CAZAC-agnostic and transfers verbatim to a second example, the ellipsope  $\mathcal{E}_n = \{X \succeq 0 : X_{ii} = 1\}$  — Felzenszwalb–Klivans–Paul’s (2021) own worked case, where their mechanism stalls at set-level convergence; we prove single-point convergence there unconditionally. Their own Theorem 20 further settles, by direct citation, that rank-1 points are exactly the fixed points whose full neighborhood converges to them; a 140-trial sweep finds no exceptions, and whether the exceptional set has measure zero remains open (Q7). Theorem B (unconditional, obstruction): non-CAZAC fixed points exist and some are exposed, so no deterministic selection rule can guarantee convergence to the CAZAC set — only a probabilistic statement over initializations is possible. We close with the open-problems ledger (Q1 settled here as Theorem A; Q2–Q7 open or partially resolved) and a tiered accounting of exactly which claims are proved, proved conditionally, numerically verified, or conjectural.

## 1 Setup

Fix  $n \geq 1$  and  $N \geq 1$ . Let  $F$  denote the unnormalized discrete Fourier transform on  $\mathbb{C}^n$ ,  $F(u)(k) = \sum_m u(m)e^{-2\pi i k m/n}$ , so Parseval’s identity reads  $\|Fu\|^2 = n\|u\|^2$ . The state space is

$x = (x_0, \dots, x_{N-1}) \in (\mathbb{C}^n)^N \cong \mathbb{R}^{2Nn}$ , with real inner product  $\langle u, v \rangle_{\mathbb{R}} = \operatorname{Re} \langle u, v \rangle$  and induced norm  $\|x\|^2 = \sum_{j,k} |x_j(k)|^2$ . Write the *coupled spectrum*  $S_x(k) = \sum_{j=0}^{N-1} |F(x_j)(k)|^2$ .

**Definition 1.1** (Coupled CAZAC system).  $x$  is a coupled CAZAC system if  $|x_j(k)| = 1$  for all  $j, k$  and  $S_x(k) = Nn$  for all  $k$ . Write  $\operatorname{CAZ}(n, N)$  for this set; for  $N = 1$  this is the classical CAZAC condition.  $\operatorname{CAZ}(n, N)$  is nonempty for every  $n \geq 1, N \geq 1$ : a Zadoff–Chu sequence exists for every length  $n$  [8], and  $N$  copies of one satisfy the coupled constraint.

For  $\tau \geq 1$ , the convex relaxation is

$$C_\tau = \{x \in (\mathbb{C}^n)^N : |x_j(k)| \leq \tau \forall j, k, \quad S_x(k) \leq Nn \forall k\}.$$

$C_\tau$  is compact (amplitude bounds), convex, semialgebraic, and second-order-cone representable (each constraint is a Euclidean-norm bound on a linear image of  $x$ ); it is invariant under global phase rotation of each  $x_j$ , cyclic shifts, and (for the coupled constraint) any transformation preserving  $S_x$ .

**Iteration.** Given  $x^0 \neq 0$ , the algorithm is *iterated linear optimization*:

$$x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle x^\ell, x \rangle_{\mathbb{R}}. \quad (\star)$$

The argmax may be non-unique;  $(\star)$  permits any measurable selection. The paper’s practical schedule presolves at  $\tau = 2 + 1/\ell$  for  $\ell = 1, \dots, 10$ , then fixes  $\tau = 1$  for the main phase; all theorems below concern the main phase at fixed  $\tau \geq 1$  (in particular  $\tau = 1$ ), and hold for any measurable selection rule, so the finitely many presolve steps do not affect any conclusion.

**Definition 1.2** (Fixed point).  $x^*$  is a fixed point of  $(\star)$  if  $x^* \in \arg \max_{x \in C_\tau} \langle x^*, x \rangle_{\mathbb{R}}$ , equivalently  $\sigma_{C_\tau}(x^*) = \|x^*\|^2$ , where  $\sigma_C(r) = \max_{x \in C} \langle r, x \rangle_{\mathbb{R}}$  is the support function of  $C$ .

Three increasingly strong convergence questions organize everything that follows: (1) does  $x^\ell$  converge to a single point, not merely have limit points; (2) is that limit CAZAC; (3) at what rate.

## 2 The iteration as a generalized power method

**Proposition 2.1.** With  $f(x) = \frac{1}{2}\|x\|^2$  (convex,  $\nabla f(x) = x$ ),  $(\star)$  is exactly  $x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle \nabla f(x^\ell), x \rangle_{\mathbb{R}}$ : the conditional-gradient / generalized-power-method update for maximizing the convex function  $f$  over the compact convex set  $C_\tau$  [5], also called *iterated linear optimization* [3].

*Proof.* Immediate from the definitions. □

**Proposition 2.2** (Max-norm points are exactly CAZAC points). For every  $x \in C_\tau$  ( $\tau \geq 1$ ),  $\|x\|^2 \leq Nn$ . For  $\tau = 1$ , equality holds iff  $x \in \operatorname{CAZ}(n, N)$ . The maximum is attained for every  $n, N, \tau$ .

*Proof.* By Parseval,  $\|x\|^2 = \sum_j \|x_j\|^2 = \frac{1}{n} \sum_k S_x(k) \leq \frac{1}{n} \cdot n \cdot Nn = Nn$ , with equality iff  $S_x(k) = Nn$  for every  $k$ . Separately, at  $\tau = 1$ ,  $\|x\|^2 = \sum_{j,k} |x_j(k)|^2 \leq Nn$  with equality iff every  $|x_j(k)| = 1$ . Both chains tight simultaneously forces both CAZAC conditions; conversely a CAZAC system attains  $Nn$ . Attainment:  $N$  copies of a Zadoff–Chu sequence of length  $n$  lie in  $C_1 \subseteq C_\tau$  with  $\operatorname{norm}^2 = Nn$ . □

This reframes the nonconvex feasibility problem “find a CAZAC system” as the convex maximization  $\max_{x \in C_1} \|x\|^2$ , whose optimal value  $Nn$  is known a priori: an iterate is  $\delta$ -suboptimal iff  $Nn - \|x^\ell\|^2 = \delta$ .

**Proposition 2.3** (Monotone ascent, square-summable increments). *Along the iteration at fixed  $\tau$ , with  $x^\ell \neq 0$  and  $x^0 \in C_\tau$ :*

- (1)  $\langle x^\ell, x^{\ell+1} \rangle_{\mathbb{R}} \geq \|x^\ell\|^2$ ;
- (2)  $\|x^{\ell+1}\| \geq \|x^\ell\|$ , and  $\|x^\ell\| \uparrow \rho \leq \sqrt{Nn}$ ;
- (3)  $\|x^{\ell+1} - x^\ell\|^2 \leq \|x^{\ell+1}\|^2 - \|x^\ell\|^2$ , hence  $\sum_{\ell \geq 0} \|x^{\ell+1} - x^\ell\|^2 \leq Nn - \|x^0\|^2$  and  $\|x^{\ell+1} - x^\ell\| \rightarrow 0$ .

*Proof.* Parts (1)–(2) and the bound in (3) instantiate the argument of [3]’s Theorem 1 for this  $C_\tau$ ; only the explicit telescoped constant  $Nn - \|x^0\|^2$  in (3) is not in their statement, since it uses Prop. 2.2’s CAZAC-specific bound  $Nn$  in place of a general compactness bound. (1)  $x^\ell \in C_\tau$  is feasible for the SOCP whose optimizer is  $x^{\ell+1}$ , so the optimum is at least  $\langle x^\ell, x^\ell \rangle_{\mathbb{R}} = \|x^\ell\|^2$ ; this is where  $x^\ell \in C_\tau$  is used, which is why the hypothesis is stated for  $x^0$  and not just  $x^{\ell \geq 1}$  (every  $x^{\ell+1}$  for  $\ell \geq 0$  is automatically in  $C_\tau$ , being an argmax over  $C_\tau$ ). (2) Cauchy–Schwarz:  $\|x^\ell\| \|x^{\ell+1}\| \geq \langle x^\ell, x^{\ell+1} \rangle_{\mathbb{R}} \geq \|x^\ell\|^2$ ; divide by  $\|x^\ell\|$ ; monotone and bounded above by Prop. 2.2. (3)  $\|x^{\ell+1} - x^\ell\|^2 = \|x^{\ell+1}\|^2 + \|x^\ell\|^2 - 2\langle x^\ell, x^{\ell+1} \rangle_{\mathbb{R}} \leq \|x^{\ell+1}\|^2 - \|x^\ell\|^2$  by (1); telescope and apply (2).  $\square$

The hypothesis  $x^0 \in C_\tau$  costs nothing: the paper’s own presolve schedule (Section 1) hands the main phase a point in  $C_{2.1} \setminus C_1$ , not in  $C_1$ , but every  $x^{\ell+1}$  for  $\ell \geq 0$  is by construction an argmax over  $C_\tau$  and hence lies in  $C_\tau$  automatically — so re-indexing the main phase to start from its own *first* iterate (which is in  $C_\tau$  regardless of where the presolve output sat) restores the hypothesis with no loss; every result below that depends on Prop. 2.3 inherits this same one-step re-indexing.

Square-summable increments alone do not imply convergence of the full sequence (harmonic-type sums can wander); closing exactly this gap is Theorem A.

**Proposition 2.4** (Limit points are fixed points; CAZAC points are fixed points). *Every limit point of  $(x^\ell)$  is a fixed point of  $(\star)$ . Every  $x^* \in \text{CAZ}(n, N)$  is a fixed point of the  $\tau = 1$  iteration, and moreover a global maximizer of its own functional:  $\arg \max_{x \in C_1} \langle x^*, x \rangle_{\mathbb{R}} \ni x^*$  with optimal value  $Nn$ .*

*Proof.* Let  $x^{\ell_j} \rightarrow \bar{x}$ . By Prop. 2.3(3),  $x^{\ell_j+1} \rightarrow \bar{x}$  as well.  $\sigma_{C_\tau}$  is continuous (finite convex on  $\mathbb{R}^{2Nn}$ ), and  $\langle x^{\ell_j}, x^{\ell_j+1} \rangle_{\mathbb{R}} = \sigma_{C_\tau}(x^{\ell_j})$  by definition of the update; pass to the limit. For the second claim, any  $y \in C_1$  satisfies  $\langle x^*, y \rangle_{\mathbb{R}} \leq \|x^*\| \|y\| \leq \sqrt{Nn} \sqrt{Nn} = Nn = \|x^*\|^2$  by Cauchy–Schwarz and Prop. 2.2, with equality at  $y = x^*$ .  $\square$

The same Cauchy–Schwarz argument shows any maximal-norm point of *any* compact convex set is a fixed point of iterated linear optimization; nothing CAZAC-specific is used beyond Prop. 2.2 itself — a fact that drives the ellipsope corollary in §3.1.

### 3 Theorem A — global convergence

**Theorem 3.1** (Theorem A — Proved, unconditional). *Fix  $\tau \geq 1$ . For any  $x^0 \neq 0$  and any measurable selection  $x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle x^\ell, x \rangle_{\mathbb{R}}$ , the sequence  $(x^\ell)$  has finite length,  $\sum_\ell \|x^{\ell+1} - x^\ell\| < \infty$ , and converges to a single fixed point  $\bar{x} \in C_\tau$ .*

*Proof.* Apply the abstract descent theorem of Attouch, Bolte, and Svaiter [1, Thm. 2.9] to  $F(x) = -\frac{1}{2}\|x\|^2 + \iota_{C_\tau}$  (so minimizing  $F$  is maximizing  $\|x\|^2/2$  on  $C_\tau$ ):

- **Sufficient decrease** ( $a = \frac{1}{2}$ ): exactly Prop. 2.3(3).

- **Relative error** ( $b = 1$ ): first-order optimality of the argmax step gives  $x^\ell \in N_{C_\tau}(x^{\ell+1})$ , the normal cone at the new iterate. Since  $\partial F(x^{\ell+1}) = -x^{\ell+1} + N_{C_\tau}(x^{\ell+1})$  (sum rule for  $C^1$  plus indicator of a convex set), the vector  $w = x^\ell - x^{\ell+1}$  lies in  $\partial F(x^{\ell+1})$ , and  $\|w\| = \|x^{\ell+1} - x^\ell\|$  *exactly*: the subgradient residual is literally the step. This identity is specific to  $f = \frac{1}{2}\|x\|^2$ ; for a generic convex objective the relative-error condition would be a genuine additional hypothesis, but here it is free. (The identical construction, for a general smooth strongly convex objective where it holds only as a Lipschitz-constant inequality rather than an exact equality, is Aktas–Kroer’s [4] Lemma 3.1 eq. (27)–(28); our case is the special instance where that Lipschitz constant is exactly 1, so the inequality becomes an equality — this observation is a corollary of their general lemma, not an independent discovery.)
- **Compactness/continuity**: iterates lie in compact  $C_\tau$ ;  $F$  is continuous there.
- **KL property**:  $C_\tau$  is defined by polynomial (in)equalities and  $F$  is polynomial plus the indicator of a semialgebraic set, hence semialgebraic, hence KL [2].

Critical points of  $F$  are exactly the fixed points of  $(\star)$ :  $0 \in \partial F(x^*)$  iff  $x^* \in N_{C_\tau}(x^*)$  (the sum rule above) iff  $x^*$  maximizes  $\langle x^*, \cdot \rangle_{\mathbb{R}}$  over  $C_\tau$ , which is the Fixed-point Definition of Section 1. The theorem then gives finite length and convergence to a single critical point.  $\square$

The theorem is silent on *which* fixed point the limit is; that is exactly where Theorem B enters. Selection-independence, and the fact that the presolve is finitely many extra solves, mean the conclusion applies verbatim to any concrete implementation of  $(\star)$ .

### 3.1 A second worked example: the ellipsope

The proof of Theorem 3.1 uses nothing about  $C_\tau$  beyond compactness and convexity, except at the KL step, which needs semialgebraicity. Stated in general: for *any* compact convex  $C \subset \mathbb{R}^d$  and the iteration  $x^{\ell+1} \in \arg \max_{x \in C} \langle x^\ell, x \rangle$ , sufficient decrease (Prop. 2.3(3)’s proof uses only feasibility of  $x^\ell$  for the next problem and Cauchy–Schwarz) and the relative-error identity (the normal-cone argument above, using only  $x^\ell \in N_C(x^{\ell+1})$  and  $\nabla(\frac{1}{2}\|x\|^2) = x$ ) hold unconditionally; compactness gives continuity. If  $C$  is also semialgebraic, KL holds [2] and Theorem A’s conclusion — convergence to a single fixed point, not merely a connected limit set — follows for that  $C$  with no CAZAC-specific input.

We exercise this on a genuinely different  $C$ : the **ellipsope**

$$\mathcal{E}_n = \{X \in \mathbb{R}^{n \times n} : X = X^\top, X \succeq 0, X_{ii} = 1 \ \forall i\},$$

the set of  $n \times n$  correlation matrices, with iteration  $X^{\ell+1} \in \arg \max_{X \in \mathcal{E}_n} \langle X^\ell, X \rangle$  ( $\langle A, B \rangle = \text{tr}(AB)$ ). This is Felsenszwalb, Klivans, and Paul’s own example [3]: a case where they concede the fixed-point set is a continuum for  $n > 3$ , and where their own worked instance stops at set-level convergence.

**Proposition 3.2** (Proved).  $\mathcal{E}_n$  is compact, convex, and semialgebraic.

*Proof.* Convexity of the constraints is immediate. Compactness:  $X \succeq 0$  with  $X_{ii} = 1$  gives  $|X_{ij}| \leq \sqrt{X_{ii}X_{jj}} = 1$  by Cauchy–Schwarz on the associated Gram vectors, so  $\mathcal{E}_n$  is bounded, and closed as an intersection of closed sets, hence compact. Semialgebraicity: for symmetric  $X$ ,  $X \succeq 0$  is equivalent by Sylvester’s criterion to the finite conjunction of polynomial inequalities “every principal minor of  $X$  is  $\geq 0$ ”; the diagonal constraints are polynomial equalities. So  $\mathcal{E}_n$  is a basic semialgebraic set (a spectrahedron).  $\square$

**Proposition 3.3** (Proved). *Sufficient decrease, relative error, and compactness/continuity transfer verbatim from Prop. 2.3(3) and the normal-cone argument of Theorem 3.1, replacing “ $C_\tau$ ” by “ $\mathcal{E}_n$ ” and  $\langle \cdot, \cdot \rangle_{\mathbb{R}}$  by  $\text{tr}(\cdot \cdot)$ .*

*Proof.* Neither source proof uses any CAZAC structure; both use only convexity, Cauchy–Schwarz, and the normal-cone characterization of the argmax, all of which hold verbatim on  $\mathcal{E}_n \subset \mathbb{R}^{\binom{n+1}{2}}$ .  $\square$

**Proposition 3.4** (Proved). *The fixed-point set  $\mathcal{F} = \{X \in \mathcal{E}_n : X \in \arg \max_{Y \in \mathcal{E}_n} \langle X, Y \rangle\}$  of the elliptope iteration is infinite, hence (being semialgebraic) a genuine positive-dimensional continuum, not a singleton or a finite set.*

*Proof.*  $\mathcal{F}$  is semialgebraic:  $\mathcal{E}_n$  is semialgebraic (Prop. 3.2) and  $X \mapsto \sigma_{\mathcal{E}_n}(X) := \max_{Y \in \mathcal{E}_n} \langle X, Y \rangle$  is semialgebraic by the Tarski–Seidenberg theorem (a projection of a semialgebraic set), so  $\mathcal{F} = \{X \in \mathcal{E}_n : \langle X, X \rangle = \sigma_{\mathcal{E}_n}(X)\}$  is cut out by polynomial (in)equalities. A semialgebraic subset of  $\mathbb{R}^{\binom{n+1}{2}}$  that is infinite cannot be 0-dimensional (an o-minimal set with finitely many points in every cell is finite), so it suffices to exhibit infinitely many fixed points. Felzenszwalb, Klivans, and Paul [3] state exactly this:  $\mathcal{L}_n$  (their name for  $\mathcal{E}_n$ ) has infinitely many fixed points for  $n > 3$ . Concretely,  $X^* = I_n$  is one such fixed point, and its optimal face is degenerate in the strongest possible sense:  $\text{tr}(I_n X) = \sum_i X_{ii} = n$  identically on  $\mathcal{E}_n$ , so the linear functional  $\langle I_n, \cdot \rangle$  is constant there and  $\arg \max_{X \in \mathcal{E}_n} \langle I_n, X \rangle = \mathcal{E}_n$  itself — every feasible point, including every rank-1 correlation matrix  $vv^\top$ , is simultaneously optimal for  $I_n$ ’s own direction, which in particular makes  $I_n$  a fixed point with a machine-exact identity. This illustrates the phenomenon but is not itself the infinitude argument: membership of  $vv^\top$  or other points in  $\arg \max \langle I_n, \cdot \rangle$  does not make them fixed points of the map, since a fixed point must be optimal for *its own* direction, not  $I_n$ ’s; the infinitude of  $\mathcal{F}$  rests on [3]’s cited count, not on this example alone.  $\square$

**Proposition 3.5** (Verified numerically,  $n = 5$ ). *Running the iteration on  $\mathcal{E}_5$  from four starting points —  $X^0 = I_5$ ;  $I_5$  plus a small random perturbation renormalized to unit diagonal; and two independent random correlation matrices — using an interior-point SDP solver (CLARABEL via cvxpy), all four trajectories show  $\|X^{\ell+1} - X^\ell\|_F \rightarrow 0$  (to  $\sim 10^{-15}$  by iteration 15) and settle on a single matrix. Three of the four escape the fixed-point continuum entirely and converge to a rank-1 correlation matrix  $vv^\top$  (eigenvalues  $(0, 0, 0, 0, 5)$  to machine precision); the trajectory started at  $X^0 = I_5$  is an edge case in which the solver’s own deterministic tie-break returns  $I_5$  again, a valid fixed point, not a counterexample. Script: `experiments/verify_elliptope_convergence.py` in the companion repository [10]. A broader sweep (same repository, `experiments/verify_elliptope_rank_sweep.py`) of 140 random and Gaussian-perturbed trials across  $n \in \{3, 4, 5, 8, 10, 15, 20\}$  finds **every trajectory converges to rank 1**, no exceptions. This remains verified numerically only, at these tested  $n$  and trial counts, not a proof for every  $n$  or every trajectory.*

**Theorem 3.6** (Elliptope convergence — tiers as stated). *Because  $\mathcal{E}_n$  is compact, convex, and semialgebraic (Prop. 3.2), Theorem A’s general form applies to it directly: the iterated-linear-optimization sequence on the elliptope converges to a single fixed point for every initialization, **despite the fixed-point set being an infinite continuum** (Prop. 3.4) — **Proved**. Which fixed point is reached is settled only partially: the vertices (rank-1 points) are exactly the fixed points whose full neighborhood converges to them, and no other fixed point has this property, **by direct citation** to [3]’s own Theorem 20 (Remark 3.8 below). That generic nearby trajectories escape the continuum and settle specifically onto a rank-1 point is **verified numerically** (140 trials, Prop. 3.5), not established for every  $n$  or every trajectory. Whether the measure of initializations converging elsewhere is exactly zero is **open** (Conjecture, filed as Q7 below).*

*Remark 3.7* (Relation to Felzenszwalb–Klivans–Paul). FKP’s Theorem 1 (limit points exist, are fixed points, form a connected set) is the same starting point as Prop. 2.4 here. From an *arbitrary* initialization, on a fixed-point set that is a positive-dimensional continuum (Prop. 3.4), their only route to single-point convergence is an unnumbered remark that a connected *finite* fixed-point set is a singleton, which is structurally inapplicable here; a full read of the proof of their Theorem 1 confirms it only ever derives connectedness and never separately invokes finiteness. (FKP’s own Theorem 20 does give single-point convergence from a *restricted* class of initializations — any point within distance 1 of a vertex converges to it in one step, per Remark 3.8 below — but this does not cover an arbitrary starting point, which is the setting Theorem 3.6 needs.) Theorem 3.6 closes exactly this gap via the KL/finite-length route [1] rather than the connectedness-plus-finiteness route, and is therefore not subsumed by [3] even in the positive-dimensional-face case. This route itself, as noted at Theorem A, is not unique to this paper: Aktaş–Kroer’s [4] KL/Frank-Wolfe framework (built on a different but interchangeable abstract descent theorem, Bolte–Sabach–Teboulle–Vaisbourd) directly delivers the same ellipsope conclusion — every hypothesis of their Theorem 3.2 (strongly convex smooth objective, nonempty compact constraint set, KL) is verified here by Prop. 3.2, so this is a second, independent route to Theorem 3.6, not merely a plausible one. [4] does not name FKP, does not run its framework on the ellipsope itself under that name (its own Max-Cut application factorizes to a product of spheres via Burer–Monteiro, §4.3), and does not discuss positive-dimensional critical sets; the comparison to FKP should be read as “a route FKP’s own paper does not take,” not as “a route no other paper could have taken.”

*Remark 3.8* (FKP’s own Theorem 20 bears directly on Q7). Although [3] cannot prove Theorem 3.6’s single-point convergence claim, its own Theorem 20 answers a large part of Q7 (below) by direct citation, without any new argument here: *the vertices of  $\mathcal{E}_n$  (the rank-1 sign matrices  $vv^\top$ ) are exactly the **attractive** fixed points of the map  $X \mapsto \arg \max_{Y \in \mathcal{E}_n} \langle X, Y \rangle$* , in the pointwise sense that an entire neighborhood of a vertex converges to it, while every non-vertex fixed point admits (their Prop. 19) an explicit curve of nearby points whose norm strictly increases, so it cannot attract a full neighborhood either. This is a genuine dichotomy, not a numerical observation, and it is stronger than what either paper’s abstract convergence machinery gives alone: it says which fixed points *can* be limits of nearby trajectories, complementing (not overlapping) Theorem 3.6’s unconditional guarantee that *some* single limit exists. It falls short of Q7 as literally stated for a precise reason: [3]’s Prop. 19 produces, in every neighborhood of a non-vertex fixed point, *at least one* nearby point that escapes it — exactly what “not attractive” (their Def. 6) requires — but this does not show *every* nearby point escapes, which is what “repelling” (their Def. 7) would additionally require. Whether non-vertex ellipsope fixed points are repelling outright, or can be neither attractive nor repelling (as [3]’s own §3, Example 5, shows can happen in general iterated-linear-optimization systems, for an unrelated three-dimensional cone, not the ellipsope itself) is not settled by Theorem 20 either way. If some non-vertex ellipsope point is neither, Theorem 20 alone would not rule out a positive-measure set of initializations converging to it through a collective, not individually attractive, mechanism. Q7’s residual open content is exactly this measure-zero refinement, structurally the same gap as Q6.

## 4 Theorem B — obstruction: the limit need not be CAZAC

**Theorem 4.1** (Theorem B — Proved, unconditional). (a) *The fixed points of  $(\star)$  at  $\tau = 1$  with every amplitude constraint slack are exactly the points whose coupled spectral profile satisfies  $S_x(k) = Nn$  on some proper subset  $K \subsetneq \{0, \dots, n-1\}$  of frequencies and  $S_x(k) = 0$  off  $K$  (i.e.  $\hat{x}_j(k) = 0$  for every  $j$ , when  $k \notin K$ ), with time-domain modulus strictly less than 1 everywhere; such a point has  $\|x\|^2 = N|K|$ . For  $N = 1$  this forces  $|\hat{x}(k)| = \sqrt{n}$  exactly on  $K$ ;*

for  $N \geq 2$  only the coupled sum is pinned, and the individual  $|\hat{x}_j(k)|$  are otherwise free subject to  $\sum_j |\hat{x}_j(k)|^2 = Nn$  for  $k \in K$ .

- (b) There exist non-CAZAC fixed points that are the **unique** maximizer of their own linear functional over  $C_1$  — exposed points. For  $n = 2$ :  $x^* = (1, t)$ ,  $t = \sqrt{2} - 1$ , is such a point; for  $N = 2$ , its diagonal doubling  $((1, t), (1, t))$  is such a point.
- (c) Consequently no deterministic convergence-to-CAZAC theorem is possible: every measurable selection rule in  $(\star)$  retains the point of (b) exactly, once reached.

*Proof of (a), general  $N$ .* By Prop. 2.1,  $x^* \neq 0$  is a fixed point iff it is a KKT point of  $\max_y \langle x^*, y \rangle_{\mathbb{R}}$  subject to  $g_{j,m}(y) = |y_j(m)|^2 - 1 \leq 0$  (each block  $j = 1, \dots, N$ ) and the single *shared* constraint  $h_k(y) = S_y(k) - Nn \leq 0$ ; this is a convex program and Slater's condition holds at  $y = 0$ , so KKT is necessary and sufficient. The real gradients are  $\nabla_{y_j} g_{j,m}(y) = 2y_j(m)e_m$  and, since  $S_y(k) = \sum_j |\hat{y}_j(k)|^2$  couples all blocks,  $\nabla_{y_j} h_k(y) = 2F^* E_k F y_j$  for *every*  $j$  with the *same* multiplier  $\mu_k$  — this coupling is the source of the difference from  $N = 1$ . KKT at  $x^*$  reads, per block,  $x_j^* = \Lambda_j x_j^* + F^* M F x_j^*$  for diagonal  $\Lambda_j, M \geq 0$  complementary to the active sets, with  $M$  shared across  $j$ . If every amplitude constraint is slack then each  $\Lambda_j = 0$ , so  $x_j^* = F^* M F x_j^*$  for every  $j$ ; applying  $F$  and using  $FF^* = nI$  gives  $\hat{x}_j^* = nM \hat{x}_j^*$  coordinatewise for every  $j$ , i.e. for each  $k$  either  $\hat{x}_j^*(k) = 0$  for every  $j$ , or  $\mu_k = 1/n$ . Let  $K = \{k : \exists j, \hat{x}_j^*(k) \neq 0\} \neq \emptyset$ ; for  $k \in K$ , complementary slackness on the *shared* constraint  $h_k$  with  $\mu_k = 1/n > 0$  forces the active condition  $S_{x^*}(k) = Nn$  — the coupled sum, not each block's own modulus, since  $h_k$  is one constraint shared by all  $N$  blocks. Parseval (summed over blocks) gives  $\|x^*\|^2 = \frac{1}{n} \sum_k S_{x^*}(k) = \frac{1}{n} |K| \cdot Nn = N|K|$ ; if  $|K| = n$  this forces  $\|x^*\|^2 = Nn$ , hence (Prop. 2.2)  $x^* \in \text{CAZ}(n, N)$ , contradicting amplitude-slackness. So  $K \subsetneq \{0, \dots, n-1\}$ . For  $N = 1$ ,  $S_x(k) = |\hat{x}(k)|^2$ , so the coupled condition  $S_x(k) = n$  collapses to the per-block statement  $|\hat{x}(k)| = \sqrt{n}$ ; for  $N \geq 2$  the individual  $|\hat{x}_j(k)|$  are unconstrained beyond  $\sum_j |\hat{x}_j(k)|^2 = Nn$  (e.g.  $N = 2, n = 2$ :  $\hat{x}_0(0) = \sqrt{3}, \hat{x}_1(0) = 1$  satisfies  $S_x(0) = 4 = Nn$  with neither modulus equal to  $\sqrt{n} = \sqrt{2}$ ). Conversely, for such  $x^*$  and any  $y \in C_1$ , Cauchy–Schwarz per frequency gives  $\langle x^*, y \rangle_{\mathbb{R}} = \frac{1}{n} \sum_{k \in K} \sum_j \text{Re}(\hat{x}_j^*(k) \hat{y}_j(k)) \leq \frac{1}{n} \sum_{k \in K} \sqrt{S_{x^*}(k)} \sqrt{S_y(k)} \leq \frac{1}{n} \sum_{k \in K} \sqrt{Nn} \sqrt{Nn} = N|K| = \|x^*\|^2$ , attained at  $y = x^*$  — a direct support-function argument, no KKT needed for sufficiency.  $\square$

*Proof of (b),  $n = 2, N = 1$ .* For  $y = (y_0, y_1) \in \mathbb{C}^2$  feasible in  $C_1$  (so  $|y_m| \leq 1$  for each  $m$ , and  $S_y(0) = |F(y)(0)|^2 \leq 2$  directly, hence  $|F(y)(0)| \leq \sqrt{2}$ ),

$$\langle x^*, y \rangle_{\mathbb{R}} = (1-t) \text{Re } y(0) + t \text{Re } F(y)(0) \leq (1-t) + t\sqrt{2} = 1 + t^2 = \|x^*\|^2,$$

with equality iff  $y(0) = 1$  and  $F(y)(0) = \sqrt{2}$ , which forces  $y(1) = \sqrt{2} - 1 = t$ , i.e.  $y = x^*$  — a genuine separating functional exposing  $x^*$  as a vertex-like point of  $C_1$  even though  $x^* \notin \text{CAZ}(2, 1)$  ( $\|x^*\|^2 = 4 - 2\sqrt{2} < 2$ ). The  $N = 2$  doubling repeats the argument coordinatewise. Part (c) follows immediately from the existence of one such retained point.  $\square$

*Remark 4.2.* Theorem A guarantees convergence to *some* fixed point; Theorem B shows the fixed-point set genuinely contains non-CAZAC points that trap every selection rule. The remaining content of “the algorithm works” is a probabilistic/basin-of-attraction statement (Q6 below), not a deterministic one.

## 5 Theorem C — finite exact termination at regular CAZAC limits

Call  $x^* \in \text{CAZ}(n, N)$  **regular** (hypothesis (R)) if the only linear dependency among the gradients of the active constraints at  $x^*$  is the one forced by Parseval's identity; equivalently the dependency

space  $K$  (Section 6 makes this precise) has  $\dim K = 1$ . Let  $\sigma_s > 0$  denote the associated smallest singular value at a regular point.

**Lemma 5.1** (Quantitative Newton-type error bound). *Let  $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^q$  be a quadratic map (each component a polynomial of degree  $\leq 2$ ),  $\Phi(x^*) = 0$ ,  $A := D\Phi(x^*)$  surjective with smallest singular value  $\sigma > 0$ , and let  $M$  be a Lipschitz constant for  $D\Phi$  (finite and global, since  $D\Phi$  is affine). Then for every  $x$  with  $\|x - x^*\| \leq \sigma/(8M)$  and  $\|\Phi(x)\| \leq \sigma^2/(16M)$ ,*

$$\text{dist}(x, \Phi^{-1}(0)) \leq \frac{2}{\sigma} \|\Phi(x)\|.$$

*Proof.* Since  $\Phi$  is quadratic, Taylor's expansion is exact:  $\Phi(y+v) = \Phi(y) + D\Phi(y)v + B(v, v)$  with  $B$  symmetric bilinear and  $\|B(v, v)\| = \frac{1}{2} \|(D\Phi(y+v) - D\Phi(y))v\| \leq \frac{M}{2} \|v\|^2$ . Run the modified Newton iteration  $z_0 = x$ ,  $z_{i+1} = z_i - A^+ \Phi(z_i)$ , where  $A^+$  is the Moore–Penrose pseudoinverse ( $AA^+ = I_q$  since  $A$  is surjective,  $\|A^+\| = 1/\sigma$ ). Write  $b_i = \|\Phi(z_i)\|$ ,  $d_i = -A^+ \Phi(z_i)$ , so  $\|d_i\| \leq b_i/\sigma$ . Then

$$\Phi(z_{i+1}) = \Phi(z_i) - D\Phi(z_i)A^+ \Phi(z_i) + B(d_i, d_i) = (A - D\Phi(z_i))A^+ \Phi(z_i) + B(d_i, d_i),$$

so, using  $\|A - D\Phi(z_i)\| \leq M\|z_i - x^*\|$ ,

$$b_{i+1} \leq \frac{M\|z_i - x^*\|}{\sigma} b_i + \frac{M}{2\sigma^2} b_i^2. \quad (*)$$

**Claim:** every  $z_i$  satisfies  $\|z_i - x^*\| \leq \sigma/(4M)$  and  $b_i \leq b_0 2^{-i}$ . By induction: given these,  $(*)$  gives  $b_{i+1} \leq \frac{1}{4}b_i + \frac{1}{32}b_i < \frac{1}{2}b_i$  (using  $b_i \leq b_0 \leq \sigma^2/(16M)$ ). Summing the geometric increments,  $\sum_i \|d_i\| \leq (b_0/\sigma) \sum_i 2^{-i} = 2b_0/\sigma \leq \sigma/(8M)$ , so  $\|z_{i+1} - x^*\| \leq \|x - x^*\| + 2b_0/\sigma \leq \sigma/(4M)$ , closing the induction. The  $z_i$  are Cauchy with limit  $z_\infty$ ;  $b_i \rightarrow 0$  and continuity of  $\Phi$  give  $\Phi(z_\infty) = 0$ , and  $\|z_\infty - x\| \leq 2b_0/\sigma$ . No convexity is used anywhere in this argument.  $\square$

Fix  $x^* \in \text{CAZ}(n, N)$  and let  $H = \{c \in \mathbb{R}^n : \sum_k c_k = 0\}$ ,  $P_H$  the orthogonal projection onto  $H$ . Define the **balanced constraint map**  $\Phi_s(x) = (g(x), P_H h(x)/n) \in \mathbb{R}^{Nn} \times H$ , where  $g_{j,m}(x) = |x_j(m)|^2 - 1$  and  $h_k(x) = S_x(k) - Nn$  (so  $g(x) = 0$  is the amplitude constraint and  $h(x) = 0$  is the balanced spectral constraint).

**Lemma 5.2** (Zero set).  $\Phi_s^{-1}(0) = \text{CAZ}(n, N)$ , globally.

*Proof.* If  $g(x) = 0$  then  $x$  is unimodular, so by Parseval  $\sum_k h_k(x) = n\|x\|^2 - nNn = 0$ , i.e.  $h(x) \in H$ ; then  $P_H h(x) = 0$  forces  $h(x) = 0$ , so  $x \in \text{CAZ}(n, N)$ . Conversely  $\text{CAZ}(n, N) \subseteq \{g = 0, h = 0\} \subseteq \Phi_s^{-1}(0)$ .  $\square$

**Lemma 5.3** (Uniform derivative bounds).  $D\Phi_s$  is  $2\sqrt{2}$ -Lipschitz globally, and at any  $x^* \in \text{CAZ}(n, N)$ ,  $\|D\Phi_s(x^*)\|_{\text{op}} \leq 2\sqrt{N+1}$ . Both bounds are independent of  $n$ .

*Proof.* (Lipschitz.) The rows of  $Dg$  are  $2x_j(m)$ , supported on disjoint real 2-planes, so  $\|Dg(x) - Dg(y)\|_F = 2\|x - y\|$ . The  $h_k$ -row of  $Dh/n$  in block  $j$  is (the real form of)  $2\hat{x}_j(k)f_k/n$  ( $f_k(m) = e^{2\pi i k m/n}$ ), so by Parseval  $\|Dh(x)/n - Dh(y)/n\|_F^2 = 4\|x - y\|^2$ ;  $P_H$  is a contraction. Stacking: Lipschitz constant  $\leq \sqrt{4+4} = 2\sqrt{2}$ . (Operator norm at  $x^*$ .)  $Dg(x^*)Dg(x^*)^\top = 4I$  (disjoint unit 2-planes,  $|x_j^*(m)| = 1$ ); the Gram matrix of the unprojected scaled spectral rows is  $4N\delta_{kk'}$  (Parseval per block,  $S_{x^*} \equiv Nn$ ). The two blocks have operator norms 2 and  $2\sqrt{N}$ ; the stack has norm  $\leq \sqrt{4+4N} = 2\sqrt{N+1}$ .  $\square$

**Hypothesis (R) restated exactly.**  $D\Phi_s(x^*)$  is surjective onto  $\mathbb{R}^{Nn} \times H$  iff  $\dim K = 1$ , where  $K$  is the dependency space of Section 6: a pair  $(a, c)$  with  $c \in H$  annihilates the row space of  $D\Phi_s(x^*)$  iff  $\sum_{j,m} a_{j,m} Dg_{j,m} + \sum_k (c_k/n) Dh_k = 0$ , and by the definition of  $K$  these solutions are parametrized by  $c \in K$  (with  $a$  determined by  $c$ ). The Parseval solution  $c = (1, \dots, 1) \notin H$ , so the annihilator inside  $\mathbb{R}^{Nn} \times H$  has dimension  $\dim(K \cap H) = \dim K - 1$  ( $K$  always contains the constant vector, which is not in  $H$ ); hence surjectivity  $\iff \dim K = 1$ , matching hypothesis (R) as stated above. Write  $\sigma_s > 0$  for the smallest singular value of  $D\Phi_s(x^*)$  when (R) holds, and  $\rho := \sigma_s/(16\sqrt{2})$  (so  $\sigma_s/(8M) = \rho$  with  $M = 2\sqrt{2}$ , Lemma 5.3).

**Theorem 5.4** (Theorem C — Proved conditional on hypothesis (R)). *Near a regular  $x^* \in \text{CAZ}(n, N)$ :*

- (a) Sharp (linear) growth:  $D(x) \geq \frac{\sigma_s}{2\sqrt{2}} \text{dist}(x, \text{CAZ}(n, N))$  for  $x \in C_1$  near  $x^*$ , where  $D(x) = Nn - \|x\|^2$ .
- (b) KL exponent 0:  $\text{dist}(0, \partial F(x)) \geq c_0 := \sigma_s/(8\sqrt{2})$  for all non-CAZAC  $x$  in a neighborhood of  $x^*$ .
- (c) Finite exact arrival: if the Theorem A limit  $\bar{x}$  is a regular CAZAC point, then  $x^\ell = \bar{x}$  **exactly** for all sufficiently large  $\ell$ . The number of “large” steps before exact arrival is at most  $(Nn - \|x^0\|^2)/c_0^2$ .

*Proof.* (a) The objective deficit is simultaneously the  $\ell^1$  slack of both constraint families:  $D(x) = \sum_{j,m} (1 - |x_j(m)|^2) = \frac{1}{n} \sum_k (Nn - S_x(k))$ , all terms  $\geq 0$ ; in particular  $\|\Phi_s(x)\| = \sqrt{\|g(x)\|^2 + \|P_H h(x)/n\|^2} \leq \sqrt{D(x)^2 + D(x)^2} = \sqrt{2}D(x)$ . *Case 1:*  $\sqrt{2}D(x) \leq \sigma_s^2/(16M)$ ,  $M = 2\sqrt{2}$ . Apply Lemma 5.1 to  $\Phi = \Phi_s$  (quadratic; zero set  $\text{CAZ}(n, N)$  by Lemma 5.2; surjective at  $x^*$  by (R)):  $\text{dist}(x, \text{CAZ}(n, N)) \leq (2/\sigma_s)\|\Phi_s(x)\| \leq (2\sqrt{2}/\sigma_s)D(x)$ , i.e.  $D(x) \geq (\sigma_s/(2\sqrt{2}))\text{dist}(x, \text{CAZ}(n, N))$ . *Case 2:*  $\sqrt{2}D(x) > \sigma_s^2/(16M)$ , i.e.  $D(x) > \sigma_s^2/64$ . Since  $x$  is in the stated neighborhood,  $\text{dist}(x, \text{CAZ}(n, N)) \leq \|x - x^*\| \leq \rho = \sigma_s/(16\sqrt{2})$ , so  $(\sigma_s/(2\sqrt{2}))\text{dist}(x, \text{CAZ}(n, N)) \leq \sigma_s^2/64 < D(x)$ : the bound holds directly. Either case gives (a).

(b) Let  $d := \text{dist}(x, \text{CAZ}(n, N)) > 0$  and pick a nearest  $z \in \text{CAZ}(n, N)$  (exists,  $\text{CAZ}(n, N)$  compact). Any  $w \in \partial F(x)$  has the form  $w = v - x$  with  $v \in N_{C_1}(x)$  (the sum rule, as in Theorem A’s proof). Since  $z \in C_1$ , the normal-cone inequality gives  $\langle v, z - x \rangle_{\mathbb{R}} \leq 0$ , hence, using  $\|z\|^2 = Nn$ ,

$$\langle w, z - x \rangle_{\mathbb{R}} \leq -\langle x, z - x \rangle_{\mathbb{R}} = \|x\|^2 - \langle x, z \rangle_{\mathbb{R}} = -\frac{1}{2}(Nn - \|x\|^2) + \frac{1}{2}d^2 = -(F(x) - F(x^*)) + \frac{d^2}{2},$$

so Cauchy–Schwarz on the left gives  $\|w\|d \geq (F(x) - F(x^*)) - d^2/2$ . By (a) (in the form  $F(x) - F(x^*) \geq (\sigma_s/(4\sqrt{2}))d$ , i.e. (a) with  $D(x) = 2(F(x) - F(x^*))$ ), dividing by  $d$ :  $\|w\| \geq (F(x) - F(x^*))d - d/2 \geq \sigma_s/(4\sqrt{2}) - d/2$ . Since  $d \leq \|x - x^*\| \leq \rho = \sigma_s/(16\sqrt{2})$ , this gives  $\|w\| \geq \sigma_s/(4\sqrt{2}) - \sigma_s/(32\sqrt{2}) = (7/32)\sigma_s/\sqrt{2} \geq \sigma_s/(8\sqrt{2}) = c_0$ . Taking the infimum over  $w \in \partial F(x)$  proves (b).

(c) (*Full chain.*) Let  $\bar{x}$  be the Theorem A limit, regular with constants  $\rho, \sigma_s, c_0$  as in (a)/(b). Since  $x^\ell \rightarrow \bar{x}$  and  $\|x^{\ell+1} - x^\ell\| \rightarrow 0$  (Theorem A’s square-summability), fix  $L_0$  so that for all  $\ell \geq L_0$ :  $x^{\ell+1}$  lies in the (b)-neighborhood of  $\bar{x}$ , and  $\|x^{\ell+1} - x^\ell\| < c_0$ . By the exact identity from Theorem A’s proof,  $w^{\ell+1} := x^\ell - x^{\ell+1} \in \partial F(x^{\ell+1})$  with  $\|w^{\ell+1}\| = \|x^{\ell+1} - x^\ell\|$  *exactly*, at every  $\ell$  – no approximation. Suppose toward contradiction  $x^{\ell+1} \notin \text{CAZ}(n, N)$  for some  $\ell \geq L_0$ . Then  $x^{\ell+1}$  is a non-CAZAC point inside the (b)-neighborhood, so (b) gives  $\text{dist}(0, \partial F(x^{\ell+1})) \geq c_0$ . But  $w^{\ell+1} \in \partial F(x^{\ell+1})$  and  $\|w^{\ell+1}\| < c_0$  by choice of  $L_0$  — a contradiction, since  $w^{\ell+1}$  itself witnesses a subgradient of norm  $< c_0$ . Hence  $x^{\ell+1} \in \text{CAZ}(n, N)$  **exactly**: the identity is exact and (b)’s bound is a hard floor on every non-CAZAC point in the neighborhood, so a subgradient below that

floor cannot occur unless the point already lies on  $\text{CAZ}(n, N)$ . Write  $z := x^{\ell+1} \in \text{CAZ}(n, N)$ . The next step maximizes  $\langle z, y \rangle_{\mathbb{R}}$  over  $y \in C_1$ ; by Cauchy–Schwarz and Prop. 2.2 ( $\|z\| = \sqrt{Nn}$  is the global max-norm),  $\langle z, y \rangle_{\mathbb{R}} \leq \|z\| \|y\| \leq Nn$  with equality iff  $y = z$  — the argmax is the singleton  $\{z\}$ , so  $x^{\ell+2} = z$ , and by induction  $x^m = z$  for all  $m > \ell$ ; in particular  $\bar{x} = z \in \text{CAZ}(n, N)$ . The step-count bound is Theorem A’s  $\sum_{\ell} \|x^{\ell+1} - x^{\ell}\|^2 \leq Nn - \|x^0\|^2$  telescoped against  $c_0^2$  per large step (increment  $\geq c_0$ ); since every step after arrival has increment exactly 0 (the sequence is constant at  $z$ ), every large step occurs strictly before arrival, so this global count already *is* the count of large steps before arrival, with no separate argument needed to exclude post-arrival steps.  $\square$

*Remark 5.5.* This is a genuine proof by contradiction, not a heuristic squeeze: an exact identity (equality, from Theorem A) is jointly satisfiable with a hard lower bound (inequality, from (b)) on the same quantity only at  $\text{dist} = 0$ . Parts (a), (b), (c) share one status tag because they rest on exactly the same two inputs — hypothesis (R) at  $\bar{x}$  (giving  $\sigma_s, \rho$ ) and the exact H2 identity from Theorem A’s proof — and (c) adds no assumption beyond the theorem’s own hypothesis “ $\bar{x} \in \text{CAZ}(n, N)$ ” plus the unconditional singleton-argmax fact.

*Remark 5.6* (The  $N = 1$  vs.  $N = 2$  gap is local-rate-invariant). Theorem C’s constants do not distinguish  $N = 1$  from  $N = 2$ : first-order sharpness at a regular point decides the local geometry identically. The empirically observed difference in success probability between the single-sequence and coupled formulations is therefore a global (basin-of-attraction) phenomenon, not a local-rate one.

*Remark 5.7* (Novelty check against generic identifiability theory). Neither the Lewis–Luke–Malick theory of local linear convergence for averaged/alternating projections [6] nor the Drusvyatskiy–Lewis generic identifiability program [7] delivers Theorem C. The former’s core hypothesis, strongly regular (transversal) intersection, is exactly what Section 6 shows *fails* at every CAZAC point (the universal Parseval dependency), and even where satisfiable its conclusion is R-linear convergence of a different algorithm, never finite exact arrival. The latter’s genericity condition is plausibly satisfied at CAZAC points, but its machinery yields only a qualitative locally-minimal-identifiable-set statement, not a quantitative sharp-growth constant, KL exponent, or finite-step termination guarantee.

## 6 Theorem D and regularity (R)

Write  $x_u(m) = \omega^{-uT_m}$  ( $\omega = e^{2\pi i/n}$ ,  $T_m = m(m+1)/2$ ,  $n$  odd) or  $x_u(m) = \omega_{2n}^{-um^2}$  ( $\omega_{2n} = e^{i\pi/n}$ ,  $n$  even) for the Zadoff–Chu sequence of root  $u$ ,  $\gcd(u, n) = 1$  [8]. At a CAZAC tuple all constraints are active; write

$$K = \{c \in \mathbb{R}^n : \overline{x_j(m)} (F^*(c \odot \hat{x}_j))(m) \in \mathbb{R} \quad \forall j, m\}$$

for the active-gradient dependency space (the tangent space of the constraint intersection has dimension  $(N-1)n + \dim K$ ; hypothesis (R) is  $\dim K = 1$ ).

**Theorem 6.1** (Theorem D — Proved for all  $n \geq 1$ , odd and even). *Let  $n = \prod_p p^{a_p}$ ,  $d_{\max} = \prod_p p^{\lfloor a_p/2 \rfloor}$  (the largest  $d$  with  $d^2 \mid n$ ). Then for every  $n \geq 1$ , every  $N \geq 1$ , every tuple of coprime roots  $u_1, \dots, u_N$ ,*

$$K = \{c \in \mathbb{R}^n : c(k + d_{\max}) = c(k) \quad \forall k\}, \quad \dim_{\mathbb{R}} K = d_{\max},$$

*independent of  $N$ , of the roots, and of the parity of  $n$ .*

*Proof. Step 0.* A dependency  $\sum_{j,m} a_{j,m} \nabla g_{j,m} + \sum_k c_k \nabla h_k = 0$  evaluated on a perturbation is equivalent to  $F^*(c \odot \hat{x}_j)$  being coordinatewise a real multiple of  $x_j$  — the defining condition of  $K$ ; dependencies with  $c = 0$  force  $a = 0$  since amplitude gradients have disjoint supports. So the dependency space is (isomorphic to)  $K$ .

**Step 1 (self-transform).** For odd  $n$ , writing  $k = us$ ,  $\hat{x}_u(k) = G_u \omega^{uT_{u^{-1}k}}$  with  $G_u = \sum_j \omega^{-uT_j}$ ,  $|G_u|^2 = n$  by Parseval, using the exact identity  $T_{m+s} = T_m + T_s + ms$ . For even  $n$ , the parallel identity holds with  $T_m$  replaced by  $m^2$ , exponents tracked mod  $2n$ :  $\hat{x}_u(k) = G_u \omega_{2n}^{us^2}$ , using  $\omega_{2n}^{v(j+n)^2} = \omega_{2n}^{vj^2}$  (an exact integer identity valid because  $n$  is even).

**Step 2 (reduction to a correlation equation).** Fix a block with root  $u$  and write  $q_m = \overline{x_u(m)} (F^*(c \odot \hat{x}_u))(m)$ . Substituting Step 1, reindexing  $k = us \bmod n$  (a bijection;  $c'_s := c_{us}$  again real), and using the same addition-law identity as Step 1 gives, in both parities,  $q_m = G_u \sum_s c'_s \phi(s+m)$ , where  $\phi(j) = \omega^{uT_j}$  (odd  $n$ ) or  $\phi(j) = \omega_{2n}^{uj^2}$  (even  $n$ ). Since  $c'$  is real,  $q_m \in \mathbb{R}$  for all  $m$  iff  $q_m - \overline{q_m} = 0$ :

$$\sum_s c'_s e(s+m) = 0 \quad \forall m, \quad e(j) := G_u \phi(j) - \overline{G_u \phi(j)}$$

— the vanishing of a convolution against the *reality-defect* kernel  $e$ , not (a subtlety a compressed first pass can miss) mere reality of  $\sum_s c'_s \phi(s+m)$  itself, which would be a strictly stronger, false requirement:  $\phi$  alone is not conjugate-symmetric, and  $\hat{\phi}$  never vanishes (it is a pure Gauss sum of constant modulus  $\sqrt{n}$ ), so that reading would force  $\hat{c}' \equiv 0$ , contradicting  $\dim K \geq 1$ .

**Step 3 (Fourier solve).** Taking the length- $n$  DFT in  $m$  turns the convolution into  $\hat{e}(t) \hat{c}'(-t) = 0$  for every  $t$ , i.e.  $\hat{c}'(t) = 0$  wherever  $\hat{e}(-t) \neq 0$ . Writing  $\sigma = u^{-1}t$  and using Step 1's formula for  $\phi$  with  $|G_u|^2 = n$ :  $\hat{e}(t) = n(\omega^{u(\sigma-T_\sigma)} - \omega^{uT_\sigma})$  (odd  $n$ , using  $T_{-\sigma} = T_\sigma - \sigma$ ) or  $\hat{e}(t) = n(\omega_{2n}^{-u\sigma^2} - \omega_{2n}^{u\sigma^2})$  (even  $n$ ). In both parities  $\hat{e}(t) = 0$  exactly when  $\sigma^2 \equiv 0 \pmod{n}$  (odd: divide the exponent congruence by  $u$  to get  $\sigma - T_\sigma \equiv T_\sigma$ , i.e.  $\sigma \equiv 2T_\sigma = \sigma^2 + \sigma$ ; even:  $2u\sigma^2 \equiv 0 \pmod{2n} \iff u\sigma^2 \equiv 0 \pmod{n}$ , and  $u$  is a unit): the zero set is the subgroup  $Z_0 = \{t : t^2 \equiv 0 \pmod{n}\} = m_0 \mathbb{Z}_n$ ,  $|Z_0| = n/m_0 = d_{\max}$  where  $m_0 = \prod_p p^{\lceil a_p/2 \rceil}$ .

**Step 4.**  $\hat{c}'(t) = \hat{e}(u^{-1}t)$ , so “ $\text{supp } \hat{c}' \subseteq Z_0$ ” is the same condition for every root and block, giving  $K = \{c : \text{supp } \hat{c} \subseteq m_0 \mathbb{Z}_n\}$ , i.e. the  $d_{\max}$ -periodic real vectors, of real dimension  $d_{\max}$ . This count is read off directly from  $|Z_0| = d_{\max}$  and is insensitive to parity: an alternate derivation via counting conjugate-symmetric pairs  $\{t, -t \bmod n\}$  in  $Z_0$  would need to treat the self-paired frequency  $t = n/2$  separately when  $n \equiv 0 \pmod{4}$  places  $n/2 \in Z_0$ , but that bookkeeping is specific to the pair-counting route and does not arise here, since Steps 1–4 characterize  $K$  by its Fourier support directly.  $\square$

*Remark 6.2* (Scope of the written derivation). Steps 2–3 evaluate the same kind of quadratic exponential sum as Step 1's self-transform identity, via the same completion-of-square technique; the definition of the reality-defect kernel  $e$  is the one place a compressed first pass is liable to drop the conjugate term and mis-simplify the zero set (an error the schema shares structurally with the naive, incorrect transfer attempt at Björck's construction below, Lemma 6.7's own point of departure). The *conclusion*  $\dim K = d_{\max}$  is independently corroborated: by hand at  $n = 4$  (self-paired case) and  $n = 9$  (odd, non-self-paired) via direct enumeration of  $Z_0$ , and numerically across all  $21 + 43 = 64$  tested Zadoff–Chu configurations (both parities; Section 6's corollaries below cite the exact scripts). No counterexample or numerical discrepancy has been found at any tested configuration.

**Corollary 6.3.** *Hypothesis (R) holds at every Zadoff–Chu tuple iff  $n$  is squarefree — in particular at every prime  $n$ , odd or  $n = 2$ . Theorem C is therefore **unconditional** whenever the Theorem A limit lies on the symmetry orbit of a Zadoff–Chu tuple and  $n$  is squarefree.*

**Corollary 6.4.** *At Zadoff–Chu tuples with squarefree  $n$  (either parity), the CAZAC tangent dimension is exactly  $(N - 1)n + 1$ : a 1-dimensional phase circle for  $N = 1$  versus an  $(n + 1)$ -dimensional manifold for  $N = 2$ .*

Numerical verification: `experiments/verify_zc_regularity.py` for odd  $n \in \{3, 5, 7, 9, 15, 21, 25, 27, 45, 49, 75\}$  (21 configurations) and `experiments/verify_even_n_regularity.py` for the 13 even values  $n \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 24, 36, 50\}$  (43 configurations); all match  $\dim K = d_{\max}$  exactly. No Chebotarëv-type nonvanishing-minors argument is needed: the DFT of a chirp is a chirp, collapsing the dependency analysis to counting solutions of  $t^2 \equiv 0 \pmod n$ . Both scripts are in the companion repository [10].

## 6.1 The chirp-argument schema, and where it does and does not generalize

Stripped of Zadoff–Chu-specific notation, Theorem D’s proof needs exactly: (1) a closure property,  $\widehat{x_u} = G_u \cdot x_u \circ \phi_u$  for a scalar  $G_u$  and an index-set automorphism  $\phi_u$ ; (2) an algebraic addition law for the phase making the key substitution exact; (3) applying the closure twice, once to reduce the dependency condition to a correlation equation, once more to diagonalize that equation into a congruence on the Fourier support of the kernel. This replaces a linear-algebra rank question with a congruence-counting question, and is a genuinely reusable schema, not a one-off computation.

**Proposition 6.5** (Positive lead — Proved classically, newly connected here). *The Legendre symbol  $\chi$  modulo an odd prime  $p$  satisfies  $\hat{\chi}(k) = \chi(k) g(\chi)$  for  $k \neq 0$  ( $g(\chi)$  the Gauss sum,  $|g(\chi)|^2 = p$ ):  $\chi$  is, up to a scalar, its own DFT, the same closure shape used by Theorem D, with trivial reindexing.*

**Proposition 6.6** (Obstruction to this specific route — not a general impossibility result). *Cubic and higher-degree phase (“algebraic chirp”) sequences do not have Theorem D’s closure property, for the same reason completing the square is special to degree 2: the addition law  $(m + s)^d = m^d + dsm^{d-1} + \binom{d}{2}s^2m^{d-2} + \dots + s^d$  has a mixing term  $dsm^{d-1}$  of degree  $d - 1$  in  $m$ , which is linear in  $m$  (absorbable into a pure reindexing of the phase, exactly as Step 1 does) only when  $d = 2$ ; for  $d \geq 3$  and  $\gcd(d, n) = 1$ , matching the cross term  $-km$  (linear in  $m$ ) forces  $ds \equiv 0 \pmod n$ , hence  $s \equiv 0$ , so no nontrivial shift completes the  $d$ -th power the way Step 1 completes the square. Separately, quadratic Gauss sums have an exact closed-form evaluation (magnitude  $\sqrt{n}$  always); Weyl sums of degree  $\geq 3$  have no such exact evaluation, only upper bounds, so even if some other closure map existed it would not obviously carry the exact-modulus property Step 1 needs. **Together these show the specific self-transform mechanism used in Theorem D’s proof does not extend to cubic or higher phase; this is an obstruction to this route, not a proof that no route exists, and the mixing-term argument itself is stated only for  $\gcd(d, n) = 1$  — when  $d$  and  $n$  share a factor,  $ds \equiv 0 \pmod n$  can have nontrivial solutions and the argument as given does not cover that case.***

## 6.2 The Björck case: a partial result, not a proof

Björck’s construction (odd prime  $p$ , Legendre symbol  $\chi$ ) [9] defines  $b(m) = e^{i\theta(m)}$  with  $\theta$  a two-valued function of  $\chi(m)$ , differing by  $p \bmod 4$ . Unlike  $\chi$  itself,  $b$  is not literally self-dual under the DFT, but a different closure fact rescues part of the argument.

**Lemma 6.7** (Proved, exact symbolic computation).  *$b$  is exactly affine in  $\chi$ :  $b = A \cdot \mathbf{1} + B \cdot \chi + C \cdot \delta_0$  with  $C = 1 - A$ , where*

$$p \equiv 1 \pmod 4 : A = \cos \alpha, B = i \sin \alpha, \alpha = \arccos \frac{1}{1+\sqrt{p}}; \quad p \equiv 3 \pmod 4 : A = \frac{1+e^{i\beta}}{2}, B = \frac{1-e^{i\beta}}{2} = C, \beta = \alpha$$

The 3-dimensional space  $T := \text{span}_{\mathbb{C}}\{\mathbf{1}, \chi, \delta_0\}$  closes under the DFT:  $\hat{b} = C \cdot \mathbf{1} + Bg \cdot \chi + Ap \cdot \delta_0$  (using  $F(\mathbf{1}) = p\delta_0$ ,  $F(\chi) = g\chi$ ,  $F(\delta_0) = \mathbf{1}$ ,  $g = g(\chi)$  the Gauss sum,  $g = \sqrt{p}$  for  $p \equiv 1 \pmod{4}$  and  $g = i\sqrt{p}$  for  $p \equiv 3 \pmod{4}$ ).

**Lemma 6.8** (Proved). *Let  $G$  be the group of quadratic-residue units mod  $p$  (order  $q = (p-1)/2$ ), acting by decimation.  $b, \hat{b}$  are pointwise  $G$ -invariant, so  $K$  is a  $G$ -invariant subspace of  $\mathbb{R}^p$ .  $\mathbb{R}^p \cong \mathbf{1}_{\{0\}} \oplus \mathbb{R}[G] \oplus \mathbb{R}[G]$  as a  $G$ -module, with trivial-isotypic piece exactly  $T \cap \mathbb{R}^p$  (dimension 3).*

**Theorem 6.9** (Proved, both residue classes).  *$K \cap T = \text{span}\{\mathbf{1}\}$ ; i.e. restricted to the 3-dimensional symmetry-invariant candidate space, the only dependency is the Parseval one, for every prime  $p$ .*

*Proof.* Restricting the dependency condition  $\overline{b(m)} (F^*(c \odot \hat{b}))(m) \in \mathbb{R}$  for all  $m$  to  $c = \alpha \mathbf{1} + \beta \chi + \gamma \delta_0 \in T$  collapses — since  $b, \hat{b} \in T$  closes under pointwise product and  $F$  — to exactly three scalar equations, one evaluation each at  $m = 0$ ,  $m \in \text{QR}$ ,  $m \in \text{QNR}$  (the condition depends on  $m$  only through  $\chi(m) \in \{0, 1, -1\}$ ). Use  $\alpha_0 := \alpha$  and  $\beta_0 := \beta$ , the same angles fixed in Lemma 6.7 (not a phase of  $Bg$ : for  $p \equiv 1 \pmod{4}$ ,  $Bg = i \sin \alpha \cdot \sqrt{p}$  is purely imaginary, never real, so no polar-decomposition phase of  $Bg$  itself is what is meant here). The three equations reduce, after eliminating the  $m = 0$  equation (which involves only  $\alpha, \gamma$  and is solved by  $\alpha$  free), to:

$$p \equiv 1 \pmod{4} : \quad s\sqrt{p}\beta = 0, \quad -s\gamma(1 + c(p-1)) = 0 \quad (c = \cos \alpha_0 = \frac{1}{1+\sqrt{p}}, s = \sin \alpha_0);$$

$$p \equiv 3 \pmod{4} : \quad \frac{\sqrt{p}(1 - \cos \beta_0)}{2} \beta = \frac{p \sin \beta_0}{2} \gamma, \quad \frac{\sqrt{p}(1 - \cos \beta_0)}{2} \beta = -\frac{\sin \beta_0}{2} \gamma.$$

In the first case,  $\alpha_0 = \arccos \frac{1}{1+\sqrt{p}} \in (0, \pi/2)$  since  $\frac{1}{1+\sqrt{p}} \in (0, 1)$ , so  $s = \sin \alpha_0 > 0$ ;  $s\sqrt{p} \neq 0$  forces  $\beta = 0$  directly. Also  $c = \cos \alpha_0 = \frac{1}{1+\sqrt{p}} \in (0, 1)$  exactly (not merely bounded, and not  $\pm 1$ ), so  $1 + c(p-1) > 1 > 0$  for every  $p \geq 5$  (both terms of  $c(p-1)$  positive), hence  $1 + c(p-1) \neq 0$  forces  $\gamma = 0$ . In the second case,  $\beta_0 = \arccos \frac{1-p}{1+p}$  with  $\frac{1-p}{1+p} \in (-1, 1)$  strictly for every  $p \geq 3$ , so  $\beta_0 \in (0, \pi)$  and  $\sin \beta_0 > 0$ ; subtracting the two equations gives  $\gamma(\frac{p \sin \beta_0}{2} + \frac{\sin \beta_0}{2}) = 0$ , i.e.  $\gamma \sin \beta_0 (p+1)/2 = 0$ , and since  $p+1 \neq 0$  and  $\sin \beta_0 \neq 0$ ,  $\gamma = 0$ ; then  $\beta = 0$  from either equation, since the coefficient  $\sqrt{p}(1 - \cos \beta_0)/2$  is nonzero (as  $\beta_0 \neq 0$ ). In both residue classes the system forces  $\beta = \gamma = 0$  with  $\alpha$  free, giving  $K \cap T = \text{span}\{\mathbf{1}\}$  exactly.  $\square$

*Remark 6.10* (What is proved and what is not — do not upgrade). Lemma 6.7 and Lemma 6.8 and Theorem 6.9 are complete proofs. What remains **unproved** is whether  $K$  meets any of the  $q-1$  nontrivial isotypic pieces of  $\mathbb{R}^p$  under  $G$ : by Schur's lemma the dependency operator restricted to each such (complex 2-dimensional) block is a  $2 \times 2$  matrix, structurally a Jacobi-sum-type bilinear expression, and its nonvanishing was not proved. A second, independent attempt at a clean closed-form nonvanishing argument also failed: the relevant determinants are confirmed numerically nonzero at every one of 44 tested primes  $p \in [5, 200)$  (both residue classes,  $\dim K = 1$  throughout), but with no uniform lower bound of the shape a classical single-Gauss/Jacobi-sum-modulus argument would give — at  $p = 71$  the relevant gap is a factor of  $\approx 4 \times 10^3$  smaller than at the next-nearest tested prime, a genuine near-degeneracy; the script itself reports this was independently confirmed not to be a floating-point artifact by a separate 60-digit-precision recomputation at that specific prime (performed once, outside the script's own run, and recorded in its output). Script: `experiments/verify_bjorck_isotypic.py` in the companion repository [10] (the 44-prime determinant check; a separately-run set of 14 primes, all contained in this same tested range, is used for the affine-closure check of Lemma 6.7, in `experiments/verify_bjorck_regularity.py`). **The overall regularity claim at Björck points is therefore still open, not proved**, despite this genuine partial progress: what is established is a full proof restricted to the symmetry-invariant

subspace, plus strong (but non-uniform) numerical evidence, not a proof, for the complementary directions.

## 7 Open problems

**Q1 (settled).** Does  $x^\ell$  converge to a single point? Yes — Theorem 3.1.

**Q2 (open, reframed).** Characterize all fixed points of  $(\star)$  for general  $n, N$ . The stationarity system is a KKT eigenvalue equation  $x^* = (\Lambda + F^*MF)x^*$  for diagonal multipliers, structurally reminiscent of discrete prolate/Slepian time-and-band-limiting operators. Complete for  $n = 2$  (four symmetry classes); open in general. A sharper sub-question: are non-CAZAC fixed points always measure-zero repellers, or can they attract an open set of initializations?

**Q3 (partially answered).** Why does  $N = 2$  empirically always succeed where  $N = 1$  sometimes fails? The dimension lower bound  $\dim \text{CAZ}(n, N) \geq (N - 1)n + 1$  (proved) gives a mechanism, and Corollary 6.4 makes it exact at squarefree Zadoff–Chu points; but the pathological fixed-point classes are the same for  $N = 1, 2$ , so the gap is about basin measure, not existence, and remains open.

**Q4 (essentially settled locally; one remaining gap).** Rate of convergence near  $\text{CAZ}(n, N)$ : settled by Theorem C at regular points, and Theorem D now makes regularity unconditional at Zadoff–Chu points of either parity for squarefree  $n$ . Remaining: (R) at non-Zadoff–Chu CAZAC points is open beyond the Björck partial result above;  $n$ -uniformity of the true sharp-growth constant is conjectural (proved bounds scale like  $c_N/n$ , sampled ratios show no decay).

**Q5 (open, deprioritized).** What does the  $\tau$ -continuation presolve actually buy? Hypothesis: it drives the iterate onto (or near) the easier flat-spectrum variety before the main phase. Untested by ablation.

**Q6 (the central open question).** For  $N \geq 2$  and Gaussian initialization, does the iteration converge to  $\text{CAZ}(n, N)$  with probability 1 (or  $\geq 1 - c^n$ )? Requires a full fixed-point census (Q2) and a basin/measure argument. For  $N = 1$ , empirical success probability decays from  $\approx 1$  at small  $n$  to  $\approx 0.3$  at  $n = 167$ ; whether this decay is exponential or polynomial in  $n$  is unresolved.

**Q7 (partially settled by citation, residual part open).** Does every trajectory of the elliptope iteration converge to a rank-1 point? Theorem 3.6 proves single-point convergence unconditionally but not convergence to an extreme point specifically. The *pointwise* half of this question is already settled — Remark 3.8 cites [3]’s own Theorem 20: rank-1 points are exactly the fixed points whose full neighborhood converges to them, and every other fixed point admits an explicit escaping direction. What remains open is the *measure-theoretic* refinement: does the set of initializations whose trajectory converges to a non-vertex fixed point have Gaussian measure zero? [3]’s own escaping-curve argument (Prop. 19, Remark 3.8) only produces one escaping point per neighborhood, not universal escape, so it leaves open whether non-vertex elliptope points are repelling outright or could instead be collectively (not individually) attracting, the way [3]’s unrelated cone example (§3, Example 5) shows can happen in general for this style of iteration. This residual question is structurally the same conjecture as Q6 in a second, unrelated domain: if the measure is zero, it is independent evidence for Q6’s general shape; if not, CAZAC’s extra structure (the dimension count of Corollary 6.4) may be doing more work than currently credited. The numerical evidence (Prop. 3.5, 140 trials across  $n \in \{3, \dots, 20\}$ ) shows zero exceptions, consistent with — but not a proof of — the measure-zero reading. *Attack plan:* a basin/measure argument analogous to Q6’s, using the pointwise dichotomy of Theorem 20 as the starting classification of which points can even be candidates for a positive-measure limit.

## 8 Verification status and provenance

**Tiers.** *Unconditional*: complete proof, no open hypothesis. *Proved conditional on (R)*: complete proof, resting on regularity hypothesis (R). *Proved conditional on (R) and an open sub-case*: complete proof modulo one explicitly identified, unresolved lemma. *Verified numerically*: no proof; a script reproduces the claim to a stated tolerance. *Conjecture*: stated as a target, not established either way.

### Principal claims.

- Propositions 2.1–2.4 (generalized power method, max-norm = CAZAC, monotone ascent, limit points fixed): *Unconditional*.
- Theorem A (global single-point convergence): *Unconditional* as a mathematical claim, but its descent-argument mechanism is a special case of Aktaş–Kroer’s [4] general Frank-Wolfe/KL framework (2025); the paper’s own contribution here is the CAZAC-to-max  $\|x\|^2$  reduction (Prop. 2.2) that makes that template apply, not the descent argument itself — see the note at Theorem A.
- Elliptope semialgebraicity and H1–H3 transfer (Props. 3.2–3.3): *Unconditional*. Fixed-point continuum (Prop. 3.4): *Unconditional* (infinitude by citation to [3], positive-dimensionality by semialgebraicity, plus a machine-exact illustrative identity at  $I_n$ ). Single-point convergence despite that continuum (Theorem 3.6): *Unconditional* — this is a proved theorem, not a numerical finding. The pointwise vertex/non-vertex attractive dichotomy (Remark 3.8): *Unconditional*, by direct citation to [3] Theorem 20 (not a new argument here). *Separately*, that generic trajectories escape onto a rank-1 point specifically (Prop. 3.5): *Verified numerically only* (140 trials,  $n \in \{3, \dots, 20\}$ ) — this is not proved. The residual, measure-zero form of universal rank-1 convergence (Q7): *Conjecture*.
- Theorem B (obstruction, exposed non-CAZAC fixed point): *Unconditional* for every  $N$  (part (a)’s statement and proof cover general  $N$ : the coupled spectral condition  $S_x(k) = Nn$  on  $K$  is pinned, individual per-sequence moduli are not, and only  $N = 1$  collapses to a per-block modulus condition).
- Theorem C, parts (a),(b),(c), including the full contradiction-based proof of part (c): *Proved conditional on hypothesis (R)*; all three parts carry the same status tag.
- Theorem D (regularity of Zadoff–Chu points,  $\dim K = d_{\max}$ ), **both odd and even  $n$** : *Unconditional*, a complete proof for all  $n$  regardless of parity, with Steps 2–3 spelled out via the reality-defect kernel  $e$  (see the “Scope of the written derivation” remark after the proof) — independently corroborated by hand at  $n = 4, 9$  and numerically at all 64 tested configurations. Corollaries 6.3–6.4: *Unconditional* (Corollary 1 conditionally activates Theorem C, but is itself unconditional as a dimension/regularity statement).
- Legendre-symbol closure positive lead: classical fact, newly connected to the schema, *Unconditional* as stated (no claim of Björck regularity is made from it alone). Cubic/higher-phase obstruction: *proved for the specific completing-the-square mechanism* used in Theorem D’s proof, when  $\gcd(d, n) = 1$  (the mixing term becomes nonlinear in  $m$  for  $d \geq 3$ ); **not** a general impossibility result, and not addressed when  $\gcd(d, n) > 1$ .
- Björck partial result: Lemma 6.7, Lemma 6.8, Theorem 6.9 ( $K \cap T = \text{span}\{\mathbf{1}\}$ ): *Unconditional*. Full regularity (R) at Björck points: **open**, reduced to one unproved determinant-nonvanishing

lemma, *verified numerically only* at 44 primes,  $p \in [5, 200)$ . Do not read this as “proved” — the reduction is real but incomplete.

- Q1–Q7: precise open-problem statements as in Section 7; Q1 is settled (restated as a theorem above), the rest range from “partially answered” to fully open conjectures, as stated there.

**Provenance.** This document was written with substantial assistance from large language models (Claude, Anthropic) under the direction of Jeffery Kline, drawing on a longer working record of derivations kept in the companion repository [10]’s `theory/` directory; every numerical claim above is backed by a named script in that repository’s `experiments/` directory. Every claim above carries the status tag most consistent with that record; none has been upgraded in the writing of this paper.

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- [10] J. Kline, *cazac-algorithm*: companion research repository holding the `theory/` derivations, the dated `AUDIT-LEDGER.md`, and the `experiments/` scripts (including `verify_zc_regularity.py`, `verify_even_n_regularity.py`, `verify_elliptope_convergence.py`, `verify_elliptope_rank_sweep.py`, `verify_bjorck_regularity.py`, `verify_bjorck_isotypic.py`) that reproduce every numerical claim in this paper, 2026. **Not publicly hosted at time of writing** — no location is asserted here because none can currently be given; a reader cannot yet independently fetch this repository.